

Work & Energy

Building Models to Describe our World
Work, Work in nD and The Work-Energy Theorem



Outline

Introduction to Work

- What is work?

- Work as a scalar product

- Work with non-constant force: 1D

Work in nD

- 1D vs nD

- Path integrals

- Work from gravity on a pendulum

Work-Energy Theorem

- Net work

- Work-energy theorem

- Power

Introduction to Work

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Work

- The work “done” by a force, $\vec{F}$, over a displacement, $\vec{d}$, is defined to be:

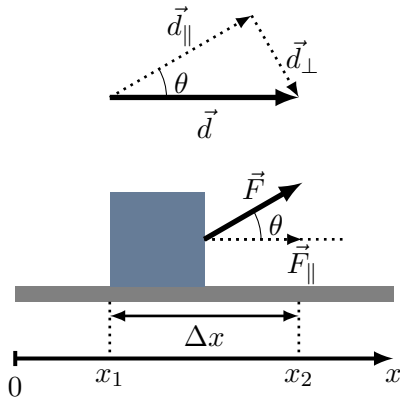
$$W = \vec{F} \cdot \vec{d} = F_x d_x + F_y d_y + F_z d_z = F d \cos \theta$$

- It is a useful quantity to connect the effect that a force has on changing the speed of an object:
 - Connecting dynamics (forces) to kinematics (change in speed).

Work as a scalar product

- Think of it as a measure of whether the force is contributing to changing the speed of the object.
- Only the component of the force parallel to the displacement can change its speed.
- The scalar product “picks out” that component of the force: $F_{\parallel} = F \cos \theta$, $d_{\parallel} = d \cos \theta$.
- Note that the *component of the force perpendicular to the displacement can change the direction of the velocity*, but not the speed.

$$\vec{F} \cdot \vec{d} = Fd \cos \theta = F_{\parallel} d = F d_{\parallel}$$

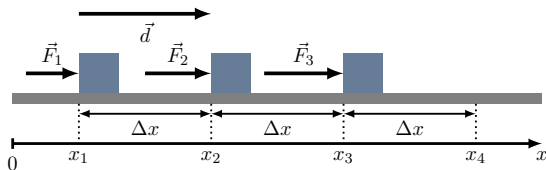


A constant force, $\vec{F}$, does work on the block over the displacement vector, $\vec{d}$.

Modeling work done by a non-constant force (1D)

- Use the component of force parallel to displacement.
- Divide the path up into segments where the force is (almost) constant.
- Add up the work done on each segment:

$$W = F_1\Delta x + F_2\Delta x + F_3\Delta x + \dots$$



A block slides along x with a force that depends on position.

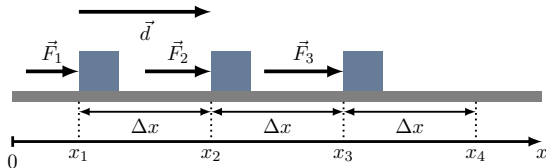
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- If the force changes continuously, $F(x)$, need to sum infinitesimally small segments:

$$W = \lim_{\Delta x \rightarrow 0} \sum F_i \Delta x = \int_{x_1}^{x_f} F(x) dx$$



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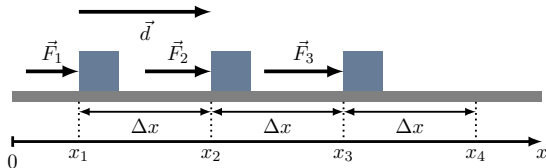
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If $F(x) = \text{constant}$:

$$W = \int_{x_1}^{x_f} F(x) dx = F \int_{x_1}^{x_f} dx = F(x_f - x_1)$$

$(x_f - x_1)$ is just the total displacement.

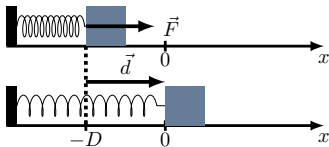
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A spring does **positive** work on a block over a total displacement, $\vec{d}$, from $x = -D$ to $x = 0$. The spring force depends on position, $F(x) = -kx$.

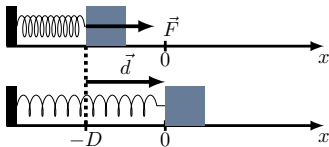
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$$\begin{aligned} W &= \int_{x_1}^{x_f} F(x) dx = \int_{-D}^0 (-kx) dx = \left[-\frac{1}{2} kx^2 \right]_{-D}^0 \\ &= \frac{1}{2} kD^2 \end{aligned}$$

The work done by a spring over a distance, D .

Work in nD

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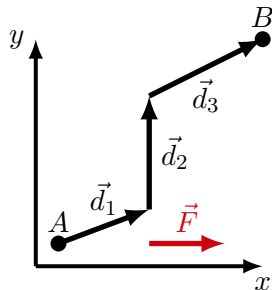
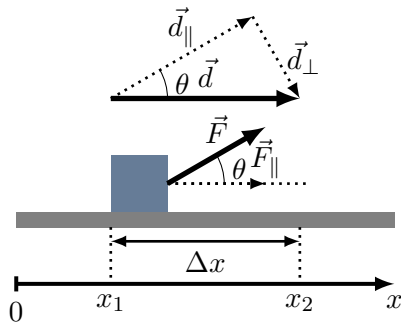
Work in ND

- Defined work, a scalar quantity, as a first step towards using scalars to describe the world instead of vectors:

$$W = \vec{F} \cdot \vec{d}$$

- The work done by a force is a measure of how much that force contributes to changing the speed of an object.
- In general, the displacement may not be one-dimensional:

$$W = \vec{F} \cdot \vec{d}_1 + \vec{F} \cdot \vec{d}_2 + \vec{F} \cdot \vec{d}_3 +$$

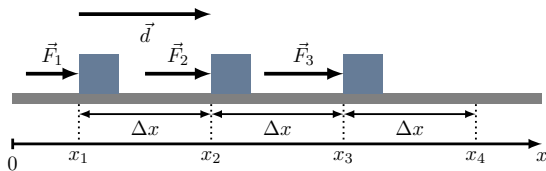


Recall 1D case

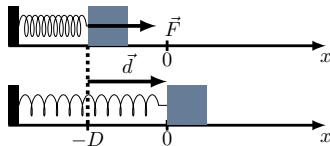
- When force changes as a function of position, we must break up the motion into infinitesimal segments and use an integral to calculate the work:

$$W = \lim_{\Delta x \rightarrow 0} \sum F_i \Delta x = \int_{x_1}^{x_f} F(x) dx$$

- The net work done by a force can depend on the path taken between x_1 and x_f :
 - For friction, it matters.
 - For a spring force, it doesn't matter.



A block slides along x with a force that depends on position.



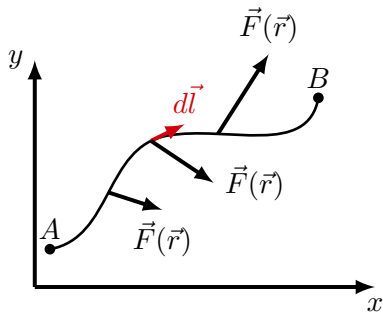
A spring does positive work on a block over a total displacement, $\vec{d}$, from $x = -D$ to $x = 0$.

In multiple dimensions

- Force can depend on position $(x,y,z) \rightarrow \vec{F} = \vec{F}(\vec{r})$, where $\vec{r}$, is the position of the object.
- A infinitesimal segment is not necessarily parallel to the x, y, z axis, use $d\vec{l}$.
- We write work as:

$$W = \int_A^B \vec{F}(\vec{r}) \cdot d\vec{l}$$

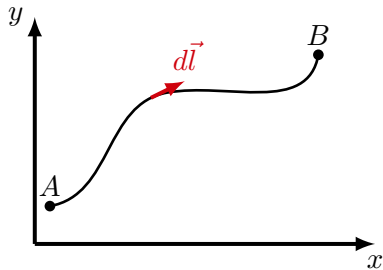
- Where $d\vec{l}$ is an infinitesimal segment, tangent to the path



A position-dependent force, $\vec{F}(\vec{r})$, does work along an arbitrary path.

Path integrals

- In general, the value of the integral will depend on what path you take between points A and B (think friction doing work).
- This is different than “normal” integrals which only depend on the end points (you just evaluate the anti-derivative at the end points).
- There are some forces (i.e. some functions) for which the work done ultimately does not depend on the path taken, like gravity, which we will discuss next week.
- **In general, it does depend on the path, and it's difficult to calculate! We want to avoid doing this integral if we can!**

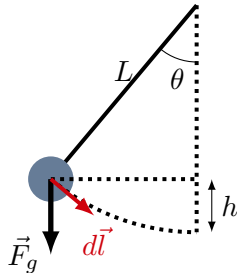


An arbitrary path over which we may need to calculate a path integral.

Example: Work done by gravity on pendulum

- We want the work done by gravity along the arc:

$$W = \int_A^B \vec{F}_g \cdot d\vec{l}$$



Gravity does work on a pendulum as it swings from $\theta = \theta_0$ to $\theta = 0$. The force is constant along the path, but the path is not a straight line.

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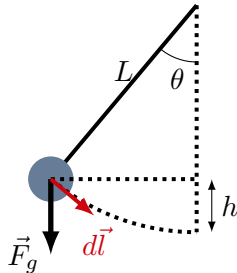
$$W = \int_A^B \vec{F}_g \cdot d\vec{l}$$

- At some angle, θ :

$$d\vec{l} = dl(\cos \theta \hat{x} - \sin \theta \hat{y})$$

$$\vec{F}_g = -mg\hat{y}$$

$$\therefore \vec{F}_g \cdot d\vec{l} = mg \sin \theta dl$$



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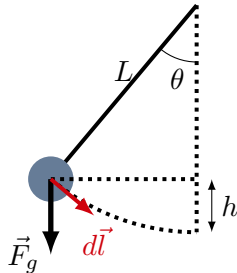
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$$\therefore \vec{F}_g \cdot d\vec{l} = mg \sin \theta dl$$

- Noting that $dl = -Ld\theta$ ($d\theta$ is in the direction of increasing θ):

$$\begin{aligned} W &= \int_A^B \vec{F}_g \cdot d\vec{l} = \int_{\theta_0}^0 -mg \sin \theta L d\theta = mgL(1 - \cos \theta_0) \\ &= mgh \end{aligned}$$



Gravity does work on a pendulum as it swings from $\theta = \theta_0$ to $\theta = 0$. The force is constant along the path, but the path is not a straight line.

Work-Energy Theorem

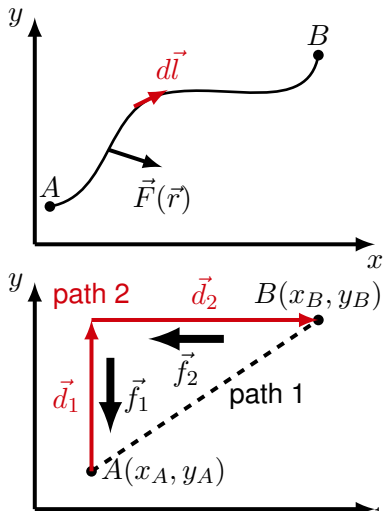
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Work, in general

$$W = \int_A^B \vec{F}(\vec{r}) \cdot d\vec{r}$$

- Force can change magnitude and direction depending on position, $\vec{F}(\vec{r})$.
- Need to define work as a “path integral”, as it depends on the path taken.
- For example, to calculate the work done by friction in moving a crate along the floor, we summed the work done on each segment (path 2):

$$W = \vec{f}_1 \cdot \vec{d}_1 + \vec{f}_2 \cdot \vec{d}_2$$



Net work done

- If multiple forces are exerted on an object, as it moves, they can each do work.
- The net work done on the object is the sum of the work done by each force exerted on the object:

$$W^{net} = \vec{F}_1 \cdot \vec{d} + \vec{F}_2 \cdot \vec{d} + \vec{F}_3 \cdot \vec{d} + \dots$$

- Very easy to show that this is equivalent to:

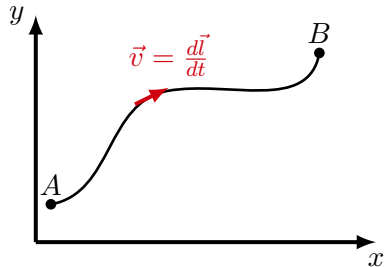
$$W^{net} = (\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots) \cdot \vec{d} = \vec{F}^{net} \cdot \vec{d}$$

That is, the net work done on the object is the same as the work done by the net force.

Work-Energy theorem

- In general, the **net** work done on an object can be written as:

$$W^{net} = \int_A^B \vec{F}^{net}(\vec{r}) \cdot d\vec{l}$$



Instantaneous velocity, $\vec{v}$ is parallel to displacement, $d\vec{l}$.

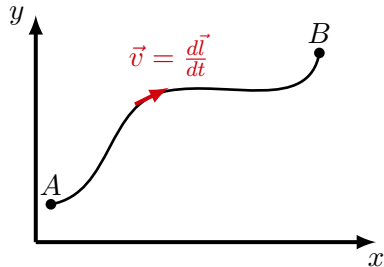
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$$W^{net} = \int_A^B \vec{F}^{net}(\vec{r}) \cdot d\vec{l}$$

- Using Newton's Second Law:

$$F^{net} = m\vec{a} = m \frac{d\vec{v}}{dt}$$
$$W^{net} = \int_A^B m \frac{d\vec{v}}{dt} \cdot d\vec{l}$$



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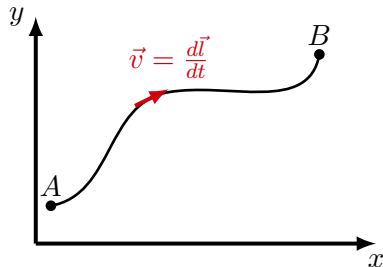
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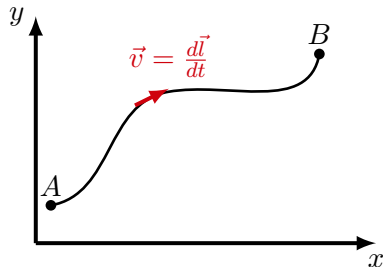
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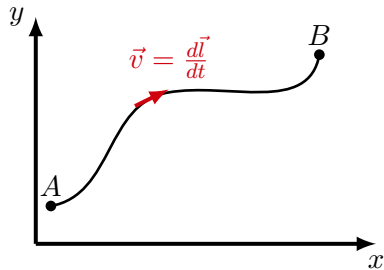
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Instantaneous velocity, $\vec{v}$ is parallel to displacement, $d\vec{l}$.

Work-Energy Theorem

- Introduce “Kinetic energy”:

$$K(v) = \frac{1}{2}mv^2$$

- The net work done on an object is the change in kinetic energy (final – initial)

$$W^{net} = \Delta K = K_B - K_A$$

- Now, we have a way to build models using scalar quantities (next week, we'll introduce potential energy as a way to calculate work without using force vectors or path integrals). Note the connection between dynamics (W) and kinematics (K), just like Newton's Second Law (not surprising, since this came from Newton's Second Law!).

Power

- Power is the rate at which work is being done.
- If a force does work, ΔW , in an amount of time, Δt , it did work with an average power:

$$P = \frac{\Delta W}{\Delta t}$$

- If the work done changes as a function of time, then we take the limit of $\Delta t \rightarrow 0$ to get the instantaneous power:

$$P = \frac{dW}{dt}$$

- Power is the rate at which a force changes the kinetic energy of an object (if it is the only force doing work) $\rightarrow$ think of this as the rate at which (kinetic) energy is being “given/removed” to/from the object.